Direct-Construction Pivot for $Q(x,y) = x^2 + xy + y^2$

Mechanism-Transfer Checkpoint

Offline, source-disciplined research report

2 September 2026

Abstract

We test whether the high-degree real sum-product construction can be transferred to

$$Q(x,y) = x^2 + xy + y^2 = N_{F(\omega)/F}(x - \omega y), \qquad \omega^2 + \omega + 1 = 0.$$

No unconditional power-saving set is obtained. The outcome is nevertheless sharper than a generic failure report. The literal carrier mechanism is shown to lose its multiplicative collision exactly: the maps $(g,h) \mapsto (g^2, gh, h^2)$ and $(g,h) \mapsto (g^2, h/g)$ retain $|G|^2/2$ unit-pair states. A fixed-field norm-fibre theorem rules out every polynomial-height-dense construction in one number field. Most importantly, arbitrary correlations across two good split residue fields satisfy the sharp Cartesian bound, and arbitrary correlations modulo an integer M satisfy

$$\frac{M|Q(B,B)|}{|B|^2} \geq \frac{1}{\tau(M)} = M^{-o(1)},$$

so the ordinary correlated-CRT lift cannot yield a fixed power saving. The most concrete surviving route uses traces of the norm-one family from the supplied unit-distance construction. It is reduced to one explicit full-Cartesian compression lemma, stated at the end of the report.

Source discipline. This report was prepared completely offline. It uses only the supplied frozen Phase A reproduction, the supplied unit-distance proof and guide, the frozen conclusions in the task, and the standard theorems stated explicitly in the standard-results section. No external search or verification is claimed.

Contents

1 A. Executive verdict **3**

2 B. Strongest theorems proved **3**
- 2.1 B.1. Exact carrier rigidity for the Phase A hybrid . . . 3
- 2.2 B.2. A fixed-field norm-fibre obstruction . . . 4
- 2.3 B.3. Arbitrary correlated CRT sets over integer moduli . . . 5
- 2.4 B.4. A sharp two-coordinate correlated obstruction . . . 6
- 2.5 B.5. A bounded-complexity multiplicative-container obstruction . . . 8

3 C. Conditional explicit construction **9**
- 3.1 C.1. The supplied norm-one carrier . . . 9
- 3.2 C.2. Exact control of every Cartesian pair . . . 9
- 3.3 C.3. Why the obvious trace container is not the missing proof . . . 10

4 D. Proof architecture and the Phase I transfer ledger **10**
- 4.1 D.1. Exact reconstruction of the successful sum-product mechanism . . . 10
- 4.2 D.2. Arrow-by-arrow transfer classification . . . 11
- 4.3 D.3. The exact inequality needed by the hybrid proof . . . 11

5 E. Failure audit **12**
5.1 E.1. Candidate A: direct action-carrier containers 12
5.2 E.2. Candidate B: ratio normalization 12
5.3 E.3. Candidate C: norm-compatible parametrization 12
5.4 E.4. Candidate D: cross-prime correlations 13
5.5 E.5. Separate rate ledgers 14
5.6 E.6. Eight early kill tests 14

Standard results used **15**

Supplied-source map **15**

6 F. One next unresolved statement **16**

1 A. Executive verdict

Verdict 2: a conditional construction has been reduced to one explicit lemma. There is presently no proof of an unconditional sequence $A_j \subset \mathbb{R}$ with $|Q(A_j, A_j)| \le |A_j|^{2-c}$. The explicit conditional sequence is

$$A_j = \sigma_j(\{u + u^{-1} : u \in U_j\}),$$

where U_j is the norm-one family in the supplied unit-distance construction and $\sigma_j : L_j \hookrightarrow \mathbb{R}$ is a real embedding. The exact identity

$$Q(u + u^{-1}, v + v^{-1}) = 3 + \big(1 + t(uv)\big)\big(1 + t(u/v)\big), \qquad t(w) = w + w^{-1},$$

reduces success to exponential compression of this one displayed two-input map. If the lemma in Section 6 holds with rate $\kappa > 0$, then one may take

$$c = \frac{\kappa}{2t \log 2} > 0.$$

The cycle also proves three unconditional obstruction results.

1. The Phase A unit carrier does not collapse the coefficient data of Q: it leaves exactly $|G|^2/2$ states.
2. In any fixed totally real field, polynomial-height-dense sets have $|Q(A, A)| = |A|^{2-o(1)}$.
3. Across two good split finite fields the normalized compression is at least 1, sharply; for every integer modulus, arbitrary cross-prime correlations have normalized compression at least $1/\tau(M) = M^{-o(1)}$. This closes the standard integer-CRT power-saving route, not merely its independent-product subcase.

The logical boundary is important. These results do *not* prove a universal lower bound $|Q(A, A)| \ge |A|^{2-o(1)}$ for arbitrary real sets in growing fields. They rule out precisely the mechanisms stated above.

2 B. Strongest theorems proved

2.1 B.1. Exact carrier rigidity for the Phase A hybrid

Theorem 2.1 (quadratic carrier rigidity)**.** *Let F be a field of characteristic different from 2, and let $G \subset F^\times$ be finite. The two maps*

$$\Gamma(g, h) = (g^2, gh, h^2), \qquad \Lambda(g, h) = \left(g^2, \frac{h}{g}\right)$$

have fibres of size at most 2 on G^2. If $G = -G$, every fibre has size exactly 2, and hence

$$|\Gamma(G^2)| = |\Lambda(G^2)| = \frac{|G|^2}{2}.$$

Proof. If $(g^2, gh, h^2) = (g'^2, g'h', h'^2)$, then $g' = \varepsilon g$ and $h' = \delta h$ for $\varepsilon, \delta \in \{\pm 1\}$. Equality of the middle coordinates gives $\varepsilon\delta = 1$, so (g', h') is (g, h) or $(-g, -h)$. Likewise, equality of $(g^2, h/g)$ and $(g'^2, h'/g')$ first gives $g' = \varepsilon g$, and then $h' = \varepsilon h$. If $G = -G$, simultaneous negation is always present and is fixed-point-free on G^2. □

For the frozen Phase A set

$$P = X + B^+(\varepsilon X), \qquad G = B^\times(Y), \qquad A = GP,$$

the multiplication map $G \times P \to A$ is injective, $G = -G$, and P is totally positive. Consequently

$$\left|\{(a^2, ab, b^2) : (a, b) \in A^2\}\right| = \frac{|A|^2}{2}.$$

Thus the sum-product collision $(g, h) \mapsto gh$ is completely undone, up to one global sign, when the two square terms are retained.

2.2 B.2. A fixed-field norm-fibre obstruction

For an integer $n \geq 1$, let $\tau(n)$ be the number of its positive divisors and put

$$T(Z) = \max_{1 \leq n \leq Z} \tau(n).$$

Theorem 2.2 (fixed-field height obstruction)**.** *Let F be a fixed totally real number field of degree d, let $E = F(\omega)$, and let $W_E = |\mu(E)|$. Let $\mathcal{A} \subset F$ be finite and suppose that some $\delta \in F^\times$ clears its denominators: $B = \delta\mathcal{A} \subset \mathcal{O}_F$. Define*

$$R = \max\left(2, \max_{\sigma : F \hookrightarrow \mathbb{R}} \max_{b \in B} |\sigma(b)|\right).$$

Then, for every real embedding $\iota : F \hookrightarrow \mathbb{R}$,

$$\boxed{|Q(\iota(\mathcal{A}), \iota(\mathcal{A}))| \geq \frac{|\mathcal{A}|^2}{W_E\, T((3R^2)^d)^d}.} \tag{B.1}$$

More precisely, for every nonzero $\beta \in \mathcal{O}_F$,

$$r_Q(\beta) := |\{(x, y) \in B^2 : Q(x, y) = \beta\}| \leq W_E \prod_{\substack{\mathfrak{p} \subset \mathcal{O}_F \\ \mathfrak{p} \text{ split in } E/F}} (v_\mathfrak{p}(\beta) + 1). \tag{B.2}$$

In particular, if F is fixed, $|\mathcal{A}_j| \to \infty$, and $R_j \leq |\mathcal{A}_j|^C$ for a fixed C, then

$$|Q(\mathcal{A}_j, \mathcal{A}_j)| \geq |\mathcal{A}_j|^{2 - O_{F,C}(1/\log\log|\mathcal{A}_j|)}. \tag{B.3}$$

Proof. The map

$$(x, y) \longmapsto z = x - \omega y$$

is injective because $1, \omega$ are linearly independent over the totally real field F, and $z\bar{z} = Q(x, y)$. Fix $\beta \neq 0$. Unique factorization of nonzero ideals in the Dedekind domain $\mathcal{O}_E$ shows that a solution satisfies

$$(z)\overline{(z)} = (\beta).$$

If $\mathfrak{p}\mathcal{O}_E = \mathfrak{P}\overline{\mathfrak{P}}$ and $v_\mathfrak{p}(\beta) = e$, the exponents of $\mathfrak{P}$ and $\bar{\mathfrak{P}}$ in (z) can be chosen in at most $e + 1$ ways. At an inert or ramified prime there is at most one choice. This gives the ideal factor in (B.2).

If two such elements generate the same ideal, then $z' = uz$ for a unit $u \in \mathcal{O}_E^\times$ with $u\bar{u} = 1$. Since E/F is CM, every complex embedding satisfies $|\rho(u)| = 1$. Kronecker's theorem therefore gives $u \in \mu(E)$, contributing at most W_E generators.

Let $m = |N_{F/\mathbb{Q}}(\beta)|$. Above each rational prime there are at most d prime ideals, and each local exponent is at most the corresponding exponent in m. Hence the product in (B.2) is at most $\tau(m)^d$. Since $|\sigma(\beta)| \leq 3R^2$ for every σ, one has $m \leq (3R^2)^d$. Summing all fibres, with the zero fibre containing at most the pair $(0, 0)$, proves (B.1). Scaling does not change cardinality because $Q(B, B) = \delta^2 Q(\mathcal{A}, \mathcal{A})$.

Finally, the standard maximal divisor estimate

$$T(Z) \leq \exp\left(C_\tau \frac{\log Z}{\log\log Z}\right)$$

for sufficiently large Z, with absolute C_τ, gives (B.3) when d and F are fixed. □

Corollary 2.3. *Integer intervals, fixed-field Minkowski boxes, and polynomial-height-dense subsets of any one fixed number field cannot satisfy the target with a fixed $c > 0$.*

2.3 B.3. Arbitrary correlated CRT sets over integer moduli

Definition 2.4. For $M \geq 2$, let Q_M be Q modulo M, and define

$$\kappa_Q(M) = \min_{\varnothing \neq B \subseteq \mathbb{Z}/M\mathbb{Z}} \frac{M|Q_M(B,B)|}{|B|^2}.$$

No product or independence assumption is imposed on B.

Theorem 2.5 (correlated integer-CRT capacity obstruction)**.** *For every integer $M \geq 2$,*

$$\boxed{\kappa_Q(M) \geq \frac{1}{\tau(M)}.} \tag{B.4}$$

Consequently, for every $\eta > 0$ there is a constant $c_\eta > 0$ such that

$$\kappa_Q(M) \geq c_\eta M^{-\eta} \qquad (M \geq 2). \tag{B.5}$$

In particular, there are no fixed $\delta > 0$ and moduli $M_j \to \infty$ with $\kappa_Q(M_j) \leq M_j^{-\delta}$.

Proof. For p^k, write

$$R_{\max}(p^k) = \max_t |\{(x,y) \in (\mathbb{Z}/p^k\mathbb{Z})^2 : Q(x,y) = t\}|.$$

The exact local counts are

$$R_{\max}(p^k) = \begin{cases} p^k(1 + k(1-1/p)), & p \equiv 1 \pmod 3, \\ (p+1)p^{k-1}, & p \equiv 2 \pmod 3, \text{ including } p = 2, \\ 2 \cdot 3^k, & p = 3. \end{cases} \tag{B.6}$$

Here is a proof of the three lines.

If $p \equiv 1 \pmod 3$, the two roots of $z^2 + z + 1$ lift to distinct roots modulo p^k, and an invertible linear change sends Q to $(u,v) \mapsto uv$. If $v_p(t) = j < k$, splitting j between u and v gives $(j+1)(p-1)p^{k-1}$ pairs. For $t = 0$, the count is

$$p^k + k(p-1)p^{k-1},$$

which is the maximum.

If $p \equiv 2 \pmod 3$, including $p = 2$, identify Q with the norm from the unramified quadratic extension of $\mathbb{Z}/p^k\mathbb{Z}$. The norm on units is surjective. Every admissible nonzero norm fibre has $(p+1)p^{k-1}$ elements; the zero fibre has $p^{2\lfloor k/2 \rfloor}$ elements and is no larger.

For $p = 3$, use the tamely ramified quadratic extension generated by ω. The residue-unit norm has index 2, while the norm on principal units is surjective. Every admissible nonzero norm fibre therefore has $2 \cdot 3^k$ elements; the zero fibre has 3^k elements.

Every line of (B.6) is at most $(k+1)p^k$. The Chinese remainder theorem makes a full fibre modulo $M = \prod p^k$ the product of its local fibres, so

$$R_{\max}(M) \leq M \prod_{p^k \| M} (k+1) = M\tau(M).$$

For an arbitrary correlated B, every ordered cross-pair is counted and

$$|B|^2 = \sum_{t \in Q_M(B,B)} r_B(t) \leq |Q_M(B,B)| R_{\max}(M).$$

This proves (B.4). The standard divisor bound $\tau(M) \leq C_\eta M^\eta$ gives (B.5) and the final assertion. □

Remark 2.6 (two useful sharp subcases). Over $\mathbb{F}_2$, $Q(x,y) = x \vee y$, coordinatewise. Thus on $\mathbb{F}_2^m$, $Q(b,b) = b$, so $B \subseteq Q(B,B)$ and

$$2^m |Q(B,B)| \geq |B|^2.$$

Also, for squarefree M, (B.6) gives the explicit bound

$$\kappa_Q(M) \geq 2^{-\omega(M)} = M^{-o(1)}.$$

Proposition 2.7 (additive correlations across rational primes split)**.** *Let $R = \prod_i \mathbb{Z}/p_i^{k_i}\mathbb{Z}$ with distinct p_i. Every additive subgroup $H \leq R$ is a product of its coordinate projections, and every affine coset of H splits likewise.*

Proof. The CRT coordinate idempotents are represented by ordinary integers. Since an additive subgroup is closed under integer multiplication, applying the i-th idempotent to an element of H places its i-th coordinate component back in H. Summing coordinate components gives the asserted product decomposition. Translation gives the coset statement. □

Thus linear or affine congruence conditions across distinct rational primes are not genuinely correlated; any surviving Candidate D construction would need nonlinear correlations in a setting not covered by ordinary integer-modulus lifting.

2.4 B.4. A sharp two-coordinate correlated obstruction

Theorem 2.8 (two good split residue fields)**.** *Let F and K be finite fields of respective odd cardinalities q and r. Assume that neither has characteristic 3, and that $T^2 + T + 1$ has two distinct roots in each. On $F \times K$, interpret*

$$Q\big((x,u),(y,v)\big) = \big(Q(x,y), Q(u,v)\big).$$

Then every $B \subseteq F \times K$, with no product or independence hypothesis, satisfies

$$\boxed{qr\,|Q(B,B)| \geq |B|^2.} \tag{B.8}$$

The constant is sharp: equality holds when $B = F \times K$.

We prove two local lemmas. They are included because the diagonal local inequality from the frozen input does not by itself imply either of them.

Lemma 2.9 (rectangular local inequality)**.** *Let k be a finite field of odd cardinality s, of characteristic different from 3, over which $T^2 + T + 1$ splits. Then for all $A, C \subseteq k$,*

$$s|Q(A,C)| \geq |A||C|. \tag{B.9}$$

Proof. The claim is immediate if one set is empty. Put $a = |A|$, $c = |C|$, $m = |Q(A,C)|$, and assume by symmetry that $a \leq c$. If $\rho \neq \sigma$ are the roots of $T^2 + T + 1$, the invertible change

$$u = x - \rho y, \qquad v = x - \sigma y$$

sends $Q(x,y)$ to uv. Thus every nonzero full fibre has $s - 1$ points. The zero fibre meets $A \times C$ in at most $2a$ points, since for each fixed first coordinate the equation is quadratic in the second.

If zero is absent from the image, then $ac \leq (s-1)m < sm$. Otherwise

$$ac \leq 2a + (s-1)(m-1), \tag{B.10}$$

which implies $m \geq ac/s$ whenever

$$a(2s - c) \leq s(s-1). \tag{B.11}$$

Suppose (B.11) fails. Since $a \le c$, one gets $(s-c)^2 < s$ and then

$$a + c > \frac{s(s-1)}{2s-c} + c > 2s - 2\sqrt{s}. \tag{B.12}$$

For $s \ge 19$, the last expression exceeds $(3s+1)/2$. If a nonzero value t were missing from $Q(A, C)$, all $s-1$ points of the conic $Q(x,y) = t$ would have to lie on the $s-a$ omitted vertical lines or the $s-c$ omitted horizontal lines. Each line meets the conic in at most two points, whereas (B.12) gives

$$2(s-a) + 2(s-c) < s - 1,$$

a contradiction. Hence the image is all of k, and the result follows.

The only odd split field sizes below 19 are 7 and 13. For $s = 7$, failure of (B.11) and $a+c \le 11$ leaves only $(a,c) = (5,5)$; then (B.10) forces $25 \le 10 + 6(m-1)$, so $m \ge 4 > 25/7$. For $s = 13$, failure of (B.11) and $a + c \le 20$ leaves only $(a,c) = (10,10)$; now $100 \le 20 + 12(m-1)$, so $m \ge 8 > 100/13$. If the displayed sum bounds fail, the preceding missing-conic argument already applies. This proves (B.9). □

Lemma 2.10 (symmetric weighted local inequality)**.** *Under the hypotheses of Lemma 2.9, let $w_x \ge 0$ for $x \in k$, and put*

$$M_t = \max_{Q(x,y)=t} w_x w_y,$$

with an empty-fibre maximum interpreted as zero. Then

$$s \sum_{t \in k} M_t \ge \left(\sum_{x \in k} w_x \right)^2. \tag{B.13}$$

Proof. Let χ be the quadratic character of k, extended by $\chi(0) = 0$.

If $\chi(3) = -1$, partition $k^\times$ into pairs $\{x, -x\}$. The values $Q(x,-x) = x^2$ run through the nonzero squares, while $Q(x,x) = 3x^2$ run through the disjoint nonsquares. Therefore

$$\sum_t M_t \ge w_0^2 + \sum_{\{x,-x\}} \Big(w_x w_{-x} + \max\{w_x^2, w_{-x}^2\} \Big) \ge \sum_x w_x^2.$$

Cauchy–Schwarz proves (B.13).

Suppose $\chi(3) = 1$. Write

$$a = w_0, \qquad W = \sum_{x \ne 0} w_x, \qquad C_c = \sum_{x \ne 0} w_x w_{cx} \quad (c \in k^\times),$$

and set $d(c) = 1 + c + c^2$. Let

$$P = \{c \in k^\times : \chi(d(c)) = 1\}, \qquad N = \{c \in k^\times : \chi(d(c)) = -1\}.$$

The standard quadratic-character identity $\sum_{c \in k} \chi(c^2 + c + 1) = -1$, together with the two roots of d, gives

$$|P| = \frac{s-5}{2}, \qquad |N| = \frac{s-1}{2}, \tag{B.14}$$

and $1 \in P$. Put $A_0 = \max_{c \in P} C_c$ and $E_0 = \max_{c \in N} C_c$. The slope pairs (x, cx), the pairs $(0, y)$, and the zero pair give, with the square and nonsquare outputs kept disjoint,

$$\sum_t M_t \ge a^2 + \frac{\max\{A_0, aW\} + E_0}{2}. \tag{B.15}$$

Indeed, pairing x with $-x$ loses at most a factor two in each displayed correlation sum.

Now $W^2 = \sum_{c\in k^\times} C_c$. If ω, ω^2 are the roots of d, then $C_\omega = C_{\omega^2} \le C_1 \le A_0$ by Cauchy–Schwarz. Hence (B.14) yields

$$W^2 \le \frac{s-1}{2}(A_0 + E_0). \tag{B.16}$$

If $A_0 \ge aW$, (B.15)–(B.16) give

$$s\sum_t M_t - (a+W)^2 \ge \left(\sqrt{s-1}\,a - \frac{W}{\sqrt{s-1}}\right)^2 \ge 0.$$

If $A_0 < aW$, (B.16) gives $E_0 > 2W^2/(s-1) - aW$; substituting in (B.15) gives the same square, strictly. This proves (B.13). □

Proof of Theorem 2.8. For $x \in F$, define the slice

$$B_x = \{u \in K : (x,u) \in B\}, \qquad w_x = |B_x|.$$

For each $t \in F$, choose x_t, y_t maximizing $w_x w_y$ subject to $Q(x,y) = t$, and let $S_t = \{v \in K : (t,v) \in Q(B,B)\}$. Then

$$Q(B_{x_t}, B_{y_t}) \subseteq S_t,$$

so Lemma 2.9 gives $r|S_t| \ge w_{x_t} w_{y_t} = M_t$. The sections for distinct t are disjoint, and Lemma 2.10 therefore gives

$$|Q(B,B)| = \sum_t |S_t| \ge \frac{1}{r}\sum_t M_t \ge \frac{1}{qr}\left(\sum_x w_x\right)^2 = \frac{|B|^2}{qr}.$$

For the full product, the split change of variables shows that each coordinate image is the whole field, proving sharpness. □

The proof uses a symmetric one-weight inequality, so it does not iterate formally to three or more factors. Such an iteration would require a two-weight or product-ring rectangular inequality; none is claimed here.

2.5 B.5. A bounded-complexity multiplicative-container obstruction

Theorem 2.11 (bounded union of multiplicative cosets)**.** *Let $A \subset \mathbb{R}$, $|A| \ge 2$, let $\Gamma \le \mathbb{C}^\times$ have finite rank r, and suppose*

$$A - \omega A \subseteq \bigcup_{j=1}^{M} \lambda_j \Gamma$$

for nonzero λ_j. Then

$$|A| \le M^2 C_{\mathrm{UE}}(2, 2r), \tag{B.17}$$

where $C_{\mathrm{UE}}(2,2r)$ is the uniform constant in the two-variable unit-equation theorem and depends only on r. In particular, no bounded r, M construction of this form can have $|A| \to \infty$.

Proof. The hypothesis itself implies $0 \notin A$. Fix distinct $y_0, y_1 \in A$ and set $c = \omega(y_1 - y_0) \ne 0$. For each $x \in A$, put

$$u_x = x - \omega y_0, \qquad v_x = x - \omega y_1; \qquad u_x - v_x = c.$$

Partition the values of x according to $u_x \in \lambda_i\Gamma$ and $v_x \in \lambda_j\Gamma$. In a fixed cell, write $u_x = \lambda_i\gamma$, $v_x = \lambda_j\delta$. Then

$$\frac{\lambda_i}{c}\gamma - \frac{\lambda_j}{c}\delta = 1, \qquad (\gamma,\delta) \in \Gamma^2.$$

The two-variable unit-equation theorem bounds the number of solutions by a constant depending only on the rank $2r$ of Γ^2. There are M^2 cells. □

3 C. Conditional explicit construction

3.1 C.1. The supplied norm-one carrier

The unit-distance proof supplies the following fixed data and sequence. There are totally real fields L_j of degrees $f_j \to \infty$, CM fields $K_j = L_j(i)$, fixed rational primes $q_1, \dots, q_t$, and a constant H such that

$$h(K_j) \le H^{f_j}, \qquad \gamma := t \log 2 - \log H > 0.$$

Writing $M_0 = q_1 \cdots q_t$, the construction gives sets

$$U_j \subseteq M_0^{-2} \mathcal{O}_{K_j}, \qquad u\bar{u} = 1, \qquad |\sigma(u)| = 1 \text{ for every complex embedding},$$

with

$$e^{\gamma f_j} \le |U_j| \le 2^{t f_j}. \tag{C.1}$$

The lower bound is the class-group-fibre count; the upper bound is the number of Boolean prime-ideal sign patterns. All constants t, H, M_0, γ are fixed as $j \to \infty$.

Define

$$t(u) = u + u^{-1} = u + \bar{u} \in L_j, \qquad T_j = t(U_j).$$

If $t(u) = t(v)$, then $(u - v)(uv - 1) = 0$, so $v = u$ or $v = u^{-1}$. Hence

$$\frac{|U_j|}{2} \le |T_j| \le |U_j|, \tag{C.2}$$

and $|T_j| \to \infty$. Moreover every real embedding of L_j sends T_j into $[-2, 2]$.

3.2 C.2. Exact control of every Cartesian pair

For arbitrary norm-one u, v, the elementary trace identities

$$t(u)^2 = t(u^2) + 2, \qquad t(u)t(v) = t(uv) + t(u/v)$$

give

$$\begin{aligned} Q(t(u), t(v)) &= t(u^2) + t(v^2) + t(uv) + t(u/v) + 4 \\ &= 3 + \big(1 + t(uv)\big)\big(1 + t(u/v)\big). \end{aligned} \tag{C.3}$$

This is an equality of full images, not an energy statement or an anchored-pair statement.

Assume the lemma in Section 6. Choose any real embedding $\sigma_j : L_j \hookrightarrow \mathbb{R}$, and set

$$A_j = \sigma_j(T_j). \tag{C.4}$$

Then σ_j is an injective field homomorphism, so

$$|A_j| = |T_j|, \qquad |Q(A_j, A_j)| = |Q(T_j, T_j)|.$$

The lemma and (C.2) give

$$|Q(A_j, A_j)| \le 4e^{-\kappa f_j} |A_j|^2.$$

Since $|A_j| \le 2^{t f_j}$,

$$e^{-\kappa f_j} \le |A_j|^{-\kappa/(t \log 2)}.$$

After discarding finitely many j to absorb the factor 4,

$$\boxed{|Q(A_j, A_j)| \le |A_j|^{2-c}, \qquad c = \frac{\kappa}{2t \log 2} > 0.} \tag{C.5}$$

Together with (C.1)–(C.2), this proves $|A_j| \to \infty$ conditionally.

3.3 C.3. Why the obvious trace container is not the missing proof

Because $U_j \subset M_0^{-2}\mathcal{O}_{K_j}$,

$$T_j \subset M_0^{-2}\mathcal{O}_{L_j}, \qquad Q(T_j, T_j) \subset M_0^{-4}\mathcal{O}_{L_j}.$$

All real conjugates of an output lie in $[0, 12]$. Clearing denominators and applying the standard Minkowski-box packing estimate gives only

$$|Q(T_j, T_j)| \le (24M_0^4 + 1)^{f_j}. \tag{C.6}$$

But $M_0 \ge 2^t$, while $|U_j|^2 \le 4^{tf_j}$. Thus the base in (C.6) is already larger than the entire trivial pair count. The conditional lemma must exploit exact collisions in (C.3); a global lattice container cannot prove it.

4 D. Proof architecture and the Phase I transfer ledger

4.1 D.1. Exact reconstruction of the successful sum-product mechanism

Let $K/\mathbb{Q}$ be totally real of degree d, with discriminant Δ_K, regulator R_K, and real embeddings $\sigma_1, \dots, \sigma_d$. The supplied frozen reproduction uses

$$B^+(T) = \{a \in \mathcal{O}_K : |\sigma_i(a)| \le T \ \forall i\},$$

$$P = X + B^+(\varepsilon X), \qquad G = B^\times(Y) = \{u \in \mathcal{O}_K^\times : |\log|\sigma_i(u)|| \le Y \ \forall i\},$$

and $A = GP$. Here $\varepsilon > 0$ is a sufficiently small absolute constant, while $X, Y \ge 2$ are in the proof-normalized regime. The exact frozen ledger is

$$(\varepsilon X)^d \Delta_K^{-1/2} \le |P| \le (2\varepsilon X + 1)^d, \tag{D.1}$$

$$\frac{Y^{d-1}}{\sqrt{d}\, R_K} \le |G| \le 10(5Y + 1)^{d-1}, \qquad |GG| \le 10(10Y + 1)^{d-1}, \tag{D.2}$$

$$|A| = |G||P|, \qquad \frac{(c_0 XY)^d}{Y\Delta_K^{3/2}} \le |A| \le (XY/c_0)^d, \tag{D.3}$$

and

$$\frac{|AA|}{|A|^2} \le \frac{|GG|}{|G|^2} \le c_0^{-d} Y^{1-d} \Delta_K^2, \tag{D.4}$$

$$\frac{|A + A|}{|A|} \le (e^Y/c_0)^d \Delta_K^{1/2}. \tag{D.5}$$

The constants $c_0 > 0$ and $\varepsilon > 0$ are absolute; they do not depend on K, d, X, Y. The equality in (D.3) follows from unit separation and positivity: a collision $g_1 p_1 = g_2 p_2$ makes the unit $g_1/g_2 = p_2/p_1$ close to 1 in every embedding, forcing it to be 1.

The field supply gives infinitely many degrees with

$$\Delta_K \le D_0^d, \qquad R_K \le \Delta_K, \tag{D.6}$$

where $D_0 > 1$ is absolute and the regulator inequality is used for $d \ge 2$. A fixed large Y turns (D.4) into $|AA| \le 2^{-d}|A|^2$. A fixed sufficiently large X pays the linear additive cost in (D.5). Finally, $|A| \le (CX)^d$ converts 2^{-d} into a fixed negative power of $|A|$.

4.2 D.2. Arrow-by-arrow transfer classification

Phase A line	**Role and rate**	**Status for Q**
Bounded-root-discriminant fields	$\Delta_K \le D_0^d$; all arithmetic costs are exponential in d.	**Transfers after modification.** High degree still converts an exponential saving into a power, but no such Q-saving is presently proved.
Additive box P	$\lvert P\rvert \ge (\varepsilon X)^d \Delta_K^{-1/2}$, with all conjugates in a short positive window.	**Geometry transfers; payoff fails.** A quadratic output has scale X^2, which cancels the X^{2d} in $\lvert A\rvert^2$.
Logarithmic unit box G	$\lvert G\rvert \gg Y^{d-1}/(\sqrt{d}R_K)$ and $\lvert GG\rvert/\lvert G\rvert^2 \ll Y^{1-d}\Delta_K^2$.	**The count transfers, the collision does not.** Theorem 2.1 leaves $\lvert G\rvert^2/2$ coefficient or scale-ratio states.
Directness $G \times P \to GP$	No multiplicity loss: $\lvert A\rvert = \lvert G\rvert\lvert P\rvert$.	**Transfers directly.** It also parametrizes every pair in A^2 exactly.
Product identity	$AA = (GG)(PP)$, so the pair (g,h) is replaced by gh.	**Fails for two moving inputs.** The terms g^2, gh, and h^2 jointly recover (g,h) up to sign.
Bounded distortion	$\lvert\sigma_i(gp)\rvert \le 2Xe^Y$, giving one additive box for $A+A$.	**Transfers only as a losing container.** It gives radius $O(X^2e^{2Y})$ for $Q(A,A)$.
Choose X after Y	A large X absorbs a fixed exponential cost relative to the *linear* benchmark $\lvert A\rvert$.	**Fails.** Relative to $\lvert A\rvert^2$, the X^2-scale cancels, and increasing X worsens any target exponent $2-c$.
Real embedding	An injective ring map preserves size, sums, and products.	**Transfers directly.** It also preserves Q because $Q \in \mathbb{Z}[x,y]$.

4.3 D.3. The exact inequality needed by the hybrid proof

Put

$$F_r(P,P) = \{p^2 + rpq + r^2q^2 : p,q \in P\}, \qquad H_r = \{g \in G : rg \in G\}.$$

Then, with $R = G/G = GG$,

$$Q(A,A) = \bigcup_{r\in R} H_r^{[2]} F_r(P,P), \qquad \sum_{r\in R} \lvert H_r^{[2]}\rvert = \frac{\lvert G\rvert^2}{2}. \tag{D.7}$$

A literal analogue of the product proof would have to show, for a fixed Y,

$$\lvert Q(GP,GP)\rvert \le \Lambda^d \lvert GG\rvert\lvert P\rvert^2 \quad \text{with} \quad \frac{\Lambda D_0^2}{c_0 Y} < 1. \tag{D.8}$$

Equation (D.7) proves that (D.8) cannot follow by merely indexing the forms by GG. It would require genuine overlap after the final addition map.

5 E. Failure audit

5.1 E.1. Candidate A: direct action-carrier containers

Let $N = |A|$, $D = A^{[2]} = \{a^2 : a \in A\}$, and $M = AA$. Since the Phase A set is symmetric and avoids zero,

$$|D| = \frac{N}{2}, \qquad |M| \geq N, \qquad Q(A, A) \subseteq D + M + D.$$

Independent component counting is therefore already at least quadratic in its own benchmark. The alternative identity

$$Q(a, b) = (a + b)^2 - ab$$

does not help: the pair $(a + b, ab)$ determines the unordered pair $\{a, b\}$, so these two coordinates retain essentially all pair information before the final subtraction.

The full archimedean container is explicit. Since $|\sigma_i(a)| \leq (1 + \varepsilon)Xe^Y$,

$$|Q(A, A)| \leq \left(6(1 + \varepsilon)^2 X^2 e^{2Y} + 1\right)^d. \tag{E.1}$$

Combining this only with the frozen lower bound for $|A|$ and $\Delta_K \leq D_0^d$ yields

$$\frac{|Q(A, A)|}{|A|^2} \leq Y^2 \left(\frac{25e^{2Y} D_0^3}{c_0^2 Y^2}\right)^d, \tag{E.2}$$

whose base is larger than 1 for $Y \geq 2$ after taking $c_0 \leq 1$. This is a failure of the available certificate, not a lower bound for the actual image.

More fundamentally, for a putative exponent $c > 0$, comparison with $|A| \leq (XY/c_0)^d$ would require

$$6(1 + \varepsilon)^2 X^2 e^{2Y} + 1 \leq (XY/c_0)^{2-c}. \tag{E.3}$$

The left side has X-degree 2, the right side degree $2 - c$. Thus the Phase A move "increase X to pay the cost" goes in the wrong direction.

5.2 E.2. Candidate B: ratio normalization

Writing $r = h/g$ gives the exact decomposition (D.7). Although $|R| = |GG|$ can be exponentially smaller than $|G|^2$, the compatible outer scales obey

$$\sum_{r \in R} |H_r| = |G|^2, \qquad \sum_{r \in R} |H_r^{[2]}| = \frac{|G|^2}{2}.$$

The union bound therefore gives only

$$|Q(A, A)| \leq \sum_{r \in R} |H_r^{[2]}| \, |F_r(P, P)| \leq \frac{|A|^2}{2}. \tag{E.4}$$

Symmetry under $(a, b) \mapsto (b, a)$ and simultaneous negation improves the constant to at most $N^2/4 + N/2$ pair orbits, but no constant factor is a fixed power saving. The small ratio set is exactly cancelled by the number of compatible outer scales.

5.3 E.3. Candidate C: norm-compatible parametrization

Theorem 2.2 rules out fixed-field, polynomial-height-dense norm constructions. Theorem 2.11 also rules out the literal idea that the entire Cartesian family $A - \omega A$ lies in a bounded union of cosets of a fixed-rank multiplicative group.

The norm-one trace carrier avoids the fixed-field theorem because the degree grows, and it has excellent archimedean control. Its available denominator container nevertheless fails by the exact comparison

$$\underbrace{(24M_0^4+1)^{f_j}}_{\text{container}} \quad \text{versus} \quad \underbrace{|U_j|^2 \le 4^{tf_j}}_{\text{all ordered pairs}}, \qquad M_0 \ge 2^t. \tag{E.5}$$

The denominator cost alone is larger than the whole pair budget. This does not disprove the trace construction; it explains why only the exact interaction identity (C.3), rather than a box count, can possibly finish it.

For comparison, a geometric progression has no hidden gain. If $A_n = \{1, R, \dots, R^{n-1}\}$ with integer $R \ge 4$, then base-R uniqueness recovers the unordered pair from

$$Q(R^i, R^j) = R^{2i} + R^{i+j} + R^{2j} \quad (i < j), \qquad Q(R^i, R^i) = 3R^{2i}.$$

Hence $|Q(A_n, A_n)| = n(n+1)/2$.

5.4 E.4. Candidate D: cross-prime correlations

Theorem 2.8 proves the sharp inequality

$$qr|Q(B,B)| \ge |B|^2$$

for every correlated $B \subseteq \mathbb{F}_q \times \mathbb{F}_r$ when both coordinates are odd good split fields. Thus correlation across two such coordinates supplies no gain at all. The one-weight step in its proof does not iterate automatically, so this sharp statement is not claimed for three or more field factors.

Theorem 2.5 includes all cross-pairs and permits arbitrary nonlinear correlations in $B \subseteq \mathbb{Z}/M\mathbb{Z}$, with any number of rational-prime coordinates. Its conclusion

$$\kappa_Q(M) \ge 1/\tau(M) = M^{-o(1)}$$

shows that no integer-modulus family can have the polynomial modular gain needed by the standard lift.

To see the rate threshold exactly, suppose hypothetically that

$$\kappa_Q(M) = \frac{M|Q_M(B,B)|}{|B|^2} \le M^{-\delta}.$$

Choose balanced integer representatives

$$A \subseteq \{-\lfloor M/2 \rfloor, \dots, \lceil M/2 \rceil - 1\}$$

of B. Then $0 \le Q(A,A) < M^2$, so each allowed residue occurs at most M times and

$$|Q(A,A)| \le M|Q_M(B,B)| = \kappa_Q(M)|A|^2 \le M^{-\delta}|A|^2.$$

Since $|A| \le M$, this would give $|Q(A,A)| \le |A|^{2-\delta}$; also $|A|^2 \ge M^{1+\delta}$, so $|A| \to \infty$. Thus polynomial modular compression would solve the real problem immediately. Theorem 2.5 proves that this premise is impossible for every integer-modulus sequence.

A gain merely exponential in the number $\omega(M)$ of distinct rational primes is not enough. For squarefree M, the elementary bound $M \ge (\omega(M)+1)!$ implies $e^{-s\omega(M)} = M^{-o(1)}$. Repeated bounded residue fields coming from three or more prime ideals in growing number fields are not covered by the sharp two-field theorem or by this integer-modulus conclusion; they remain a different, genuinely global problem.

5.5 E.5. Separate rate ledgers

Route	Cardinality gain	Output or collision term	Fatal cost or open term
Phase A	$\lvert A\rvert = \lvert G\rvert\lvert P\rvert$, exponential in d	$\lvert GG\rvert/\lvert G\rvert^2 \sim Y^{1-d}$	e^{Yd} is paid by the linear $\lvert A\rvert$ benchmark
Literal $Q(GP, GP)$	Same directness and size	Coefficient family has $\lvert G\rvert^2/2$ states	X^2 cancels; available box has e^{2Yd} cost
Trace carrier	$\lvert T_j\rvert \geq e^{\gamma f_j}/2$	Desired gain is $e^{-\kappa f_j}$	Box count pays $M_0^{4f_j}$; exact trace-pair overlap is open
Two split fields	Arbitrary correlated $B \subseteq \mathbb{F}_q \times \mathbb{F}_r$	$qr\lvert Q(B,B)\rvert \geq \lvert B\rvert^2$, sharply	No two-coordinate modular gain; no claim for three factors
Integer CRT	Density $\lvert B\rvert/M$	Correlation parameter $\kappa_Q(M)$	$\kappa_Q(M) \geq 1/\tau(M) = M^{-o(1)}$

No hidden constant in this table depends on the growing degree or modulus unless its row says so. In the trace row, t, H, M_0, γ are fixed before $f_j \to \infty$.

5.6 E.6. Eight early kill tests

Test	**Literal** $A = GP$	**Conditional trace route**
Cartesian-pair	Passed: directness parametrizes all $\lvert G\rvert^2\lvert P\rvert^2$ pairs.	Passed: (C.3) is evaluated for every $(u, v) \in U_j^2$.
Cross-pair	Fails as a compression step: different carriers leave $\lvert G\rvert^2/2$ states.	Included exactly in the unresolved trace-pair image.
Mixed-term	Not controlled by $\lvert GG\rvert$; it must be combined with both square terms.	Controlled jointly by the product in (C.3), conditional on one image bound.
Multiplicity	Passed: $G \times P \to GP$ is injective.	Passed: $u \mapsto u + u^{-1}$ is at most two-to-one.
Full-image	Component bounds give only a constant-factor orbit count.	The unresolved lemma is explicitly a full-image assertion, not an energy estimate.
Rate	Fails for the known containers by (E.2)–(E.3).	Conditional rate is explicit: $c = \kappa/(2t \log 2)$.
Arbitrarily large	Passed for the Phase A family, but without a Q-image bound.	Passed conditionally and already at the size level by (C.1)–(C.2).
Realization	Passed by any real field embedding.	Passed by $\sigma_j : L_j \hookrightarrow \mathbb{R}$.

Standard results used

For clarity, the substantial standard inputs used above are stated here in the exact forms needed.

1. **Minkowski embedding and covolume.** If K is totally real of degree d, then $a \mapsto (\sigma_1(a), \dots, \sigma_d(a))$ embeds $\mathcal{O}_K$ as a full-rank lattice in $\mathbb{R}^d$ of covolume $\sqrt{\Delta_K}$.

2. **Dirichlet's unit theorem in logarithmic form.** The image of $\mathcal{O}_K^\times/\mu(K)$ under $u \mapsto (\log|\sigma_i(u)|)_i$ is a lattice of rank $r_1 + r_2 - 1$ in the product-formula hyperplane. In the totally real normalization used in Phase A, its covolume is $\sqrt{d}R_K$.

3. **Blichfeldt's lemma.** If $\Lambda \subset \mathbb{R}^n$ is a full-rank lattice and S is bounded and measurable, some translate of S contains at least $\mathrm{vol}(S)/\operatorname{covol}(\Lambda)$ lattice points. Applied to a half-sized box and followed by taking differences, this gives the lower box counts in the frozen Phase A proof.

4. **Unique factorization of ideals.** Every nonzero ideal in the ring of integers of a number field factors uniquely into nonzero prime ideals. This is used in Theorem 2.2 to count the possible ideals (z) with $(z)\overline{(z)} = (\beta)$.

5. **Kronecker's theorem.** An algebraic integer all of whose conjugates have absolute value 1 is a root of unity. Hence the relative norm-one unit group in a CM extension of its maximal totally real subfield is finite.

6. **Divisor bound.** For every $\eta > 0$ there is C_η such that $\tau(n) \le C_\eta n^\eta$ for every $n \ge 1$. The sharper form $\max_{m \le Z} \tau(m) \le \exp(C_\tau \log Z/\log\log Z)$ for all sufficiently large Z is used in (B.3).

7. **Chinese remainder theorem.** For pairwise coprime moduli, the residue ring is the direct product of the local residue rings, and a polynomial equation modulo the product is equivalent to its coordinate equations. Thus full fibre sizes multiply.

8. **Local norm maps.** In an unramified quadratic extension of nonarchimedean local rings, the norm map on units is surjective and its finite-level kernel has the quotient-of-unit-group size used in (B.6). In a tamely ramified quadratic extension, the norm is surjective on principal units, while on residue units it is the square map. These statements give the inert and ramified lines of (B.6).

9. **Quadratic-character sum.** If k is a finite field of odd cardinality, χ is its quadratic character with $\chi(0) = 0$, and $aX^2 + bX + c$ has $a \ne 0$ and nonzero discriminant, then

$$\sum_{x \in k} \chi(ax^2 + bx + c) = -\chi(a).$$

   This is used in (B.14).

10. **Unit-equation theorem.** For fixed n and rank r, a linear equation $a_1x_1 + \cdots + a_nx_n = 1$, with nonzero coefficients and variables in a rank-r multiplicative subgroup of $(\mathbb{C}^\times)^n$, has at most $C_{\mathrm{UE}}(n, r)$ nondegenerate solutions. The constant depends only on n, r, not on the coefficients or the group. For $n = 2$, every solution is nondegenerate.

Supplied-source map

- *Phase A – Reproducing the High-Degree Sum-Product Construction, FINAL FROZEN*: definitions of P, G, A, directness, equations (D.1)–(D.5), and the exponent conversion.
- *Phase A FINAL FROZEN – Change Log*: corrected parameter ranges, the ε^d factor, the hypothesis $d \ge 2$ in $R_K \le \Delta_K$, and the fixed rescaling into the theorem parameter.

- *Planar Point Sets with Many Unit Distances* and the supplied frozen guide: the fields L_j, norm-one families U_j, class-group-fibre count, fixed denominator, bounded archimedean modulus, and exact real/complex projection mechanism used in Section C.
- The task statement: all local independent-restriction results and closed branches are treated as frozen and are not reproved.

6 F. One next unresolved statement

Trace-pair compression lemma. For some admissible unit-distance tower (L_j, K_j, U_j) described in Section C, there is a constant $\kappa > 0$, independent of j, such that for every sufficiently large j,

$$\boxed{\left|\left\{(1+t(uv))(1+t(u/v)) : u, v \in U_j\right\}\right| \le e^{-\kappa f_j}|U_j|^2, \qquad t(w) = w + w^{-1}.} \tag{F.1}$$

This is the single next statement. It controls every Cartesian cross-pair and the three terms of Q jointly. By Section C it would give the completely defined real sets (C.4) and the fixed power saving (C.5). Neither high collision energy, one norm fibre, small $|UU|$, small $|U/U|$, nor the denominator-lattice container proves (F.1).